Topic 3 – Chemical changes

Acids

Students should:		Maths skills
3.1	Recall that acids in solution are sources of hydrogen ions and alkalis in solution are sources of hydroxide ions	
3.2	Recall that a neutral solution has a pH of 7 and that acidic solutions have lower pH values and alkaline solutions higher pH values	
3.3	Recall the effect of acids and alkalis on indicators, including litmus, methyl orange and phenolphthalein	
3.4	**Recall that the higher the concentration of hydrogen ions in an acidic solution, the lower the pH; and the higher the concentration of hydroxide ions in an alkaline solution, the higher the pH**	1c
3.5	**Recall that as hydrogen ion concentration in a solution increases by a factor of 10, the pH of the solution decreases by 1**	1c
3.6	*Core Practical: Investigate the change in pH on adding powdered calcium hydroxide* or *calcium oxide to a fixed volume of dilute hydrochloric acid*	4a, 4c
3.7	**Explain the terms dilute and concentrated, with respect to amount of substances in solution**	
3.8	**Explain the terms weak and strong acids, with respect to the degree of dissociation into ions**	
3.9	Recall that a base is any substance that reacts with an acid to form a salt and water only	
3.10	Recall that alkalis are soluble bases	
3.11	Explain the general reactions of aqueous solutions of acids with: a metals b metal oxides c metal hydroxides d metal carbonates to produce salts	
3.12	Describe the chemical test for: a hydrogen b carbon dioxide (using limewater)	
3.13	Describe a neutralisation reaction as a reaction between an acid and a base	
3.14	Explain an acid-alkali neutralisation as a reaction in which hydrogen ions (H^+) from the acid react with hydroxide ions (OH^-) from the alkali to form water	

Students should:		Maths skills
3.15	Explain why, if soluble salts are prepared from an acid and an insoluble reactant: a excess of the reactant is added b the excess reactant is removed c the solution remaining is only salt and water	
3.16	Explain why, if soluble salts are prepared from an acid and a soluble reactant: a titration must be used b the acid and the soluble reactant are then mixed in the correct proportions c the solution remaining, after reaction, is only salt and water	
3.17	*Core Practical: Investigate the preparation of pure, dry hydrated copper sulfate crystals starting from copper oxide including the use of a water bath*	
3.18	Describe how to carry out an acid-alkali titration, using burette, pipette and a suitable indicator, to prepare a pure, dry salt	
3.19	Recall the general rules which describe the solubility of common types of substances in water: a all common sodium, potassium and ammonium salts are soluble b all nitrates are soluble c common chlorides are soluble except those of silver and lead d common sulfates are soluble except those of lead, barium and calcium e common carbonates and hydroxides are insoluble except those of sodium, potassium and ammonium	
3.20	Predict, using solubility rules, whether or not a precipitate will be formed when named solutions are mixed together, naming the precipitate if any	
3.21	Describe the method used to prepare a pure, dry sample of an insoluble salt	

Suggested practicals

- Carry out simple neutralisation reactions of acids, using metal oxides, hydroxides and carbonates.
- Carry out tests for hydrogen and carbon dioxide.
- Prepare an insoluble salt by precipitation.

Electrolytic processes

Students should:		Maths skills
3.22	Recall that electrolytes are ionic compounds in the molten state or dissolved in water	
3.23	Describe electrolysis as a process in which electrical energy, from a direct current supply, decomposes electrolytes	
3.24	Explain the movement of ions during electrolysis, in which: a positively charged cations migrate to the negatively charged cathode b negatively charged anions migrate to the positively charged anode	
3.25	Explain the formation of the products in the electrolysis, using inert electrodes, of some electrolytes, including: a copper chloride solution b sodium chloride solution c sodium sulfate solution d water acidified with sulfuric acid e molten lead bromide (demonstration)	
3.26	Predict the products of electrolysis of other binary, ionic compounds in the molten state	
3.27	**Write half equations for reactions occurring at the anode and cathode in electrolysis**	1c
3.28	**Explain oxidation and reduction in terms of loss or gain of electrons**	
3.29	**Recall that reduction occurs at the cathode and that oxidation occurs at the anode in electrolysis reactions**	
3.30	Explain the formation of the products in the electrolysis of copper sulfate solution, using copper electrodes, and how this electrolysis can be used to purify copper	
3.31	*Core Practical: Investigate the electrolysis of copper sulfate solution with inert electrodes and copper electrodes*	1a 4a, 4b, 4c, 4d

Suggested practicals

- Investigate the electrolysis of:
  - a copper chloride solution
  - b sodium chloride solution
  - c sodium sulfate solution
  - d water acidified with sulfuric acid
  - e molten lead bromide (demonstration).